

Task 4.6: Model Monitoring Checklist

Objective

The purpose of model monitoring is to ensure that the deployed Adult Income Prediction model continues to generate reliable and accurate predictions after deployment. Monitoring helps identify data quality issues, changes in incoming data, performance degradation, and potential fairness concerns before they significantly affect the model's predictions.

Monitoring Checklist

1. Input Data Quality

What to monitor:

- Missing values
- Invalid data types
- Unexpected or unseen categorical values
- Incorrect input formats

Alert threshold:

- More than 2% of incoming records contain validation errors.

Recommended action:

- Investigate the data source.
- Correct data preprocessing or reject invalid records until the issue is resolved.

2. Data Drift

What to monitor:

- Changes in the distribution of numerical features such as age, capital gain, and hours worked per week.
- Changes in the frequency of categorical variables such as occupation or marital status.

Alert threshold:

- Population Stability Index (PSI) greater than 0.20 or any significant distribution shift.

Recommended action:

- Investigate the cause of the drift.
- Compare the new data with the training data.
- Retrain the model if the drift persists.

3. Prediction Distribution

What to monitor:

- Percentage of predictions classified as income greater than \$50K.
- Changes in the overall prediction distribution over time.

Alert threshold:

- A change greater than $\pm 10\%$ compared to the training or historical prediction distribution.

Recommended action:

- Investigate possible data drift or changes in user behavior.
- Review the model if the change is sustained.

4. Model Confidence

What to monitor:

- Average predicted probability.
- Distribution of prediction confidence scores.

Alert threshold:

- Significant deviation from the historical average confidence.

Recommended action:

- Review model calibration.
- Check whether incoming data differs substantially from the training data.

5. Model Performance

What to monitor:

- Precision (Primary Business Metric)
- Recall
- F1-score
- ROC AUC
- PR AUC

Alert threshold:

- Precision falls below the target value of **75%**.

Recommended action:

- Evaluate the model on recent labeled data.
- Recalibrate or retrain the model if necessary.

6. Fairness Monitoring

What to monitor:

- Precision across protected groups such as sex and race.
- Differences in model performance between demographic groups.

Alert threshold:

- Precision differs by more than 10 percentage points between groups.

Recommended action:

- Perform a fairness analysis.
- Investigate potential sources of bias.
- Consider collecting more representative data or applying fairness-aware techniques if necessary.

7. Inference Performance

What to monitor:

- Average inference latency per prediction.
- System availability and prediction failures.

Alert threshold:

- Average inference time exceeds **50 milliseconds per sample**.

Recommended action:

- Optimize the inference pipeline.
- Allocate additional computational resources if required.

Retraining Strategy

The model should be retrained whenever one or more of the following conditions occur:

- The model's precision consistently falls below the target value of **75%**.
- Significant data drift is detected.
- The prediction distribution changes substantially over time.
- Fairness metrics indicate increasing disparities across demographic groups.
- A sufficient amount of newly labeled data becomes available.

Under normal operating conditions, the model should be reviewed and retrained approximately **every six months**, or earlier if any of the monitoring alerts are triggered.

Prediction Logging

For every prediction request, the following information should be recorded:

- Timestamp
- Model version
- Input validation status
- Predicted probability
- Predicted class
- Inference time

Maintaining these logs enables performance monitoring, troubleshooting, auditing, and future model evaluation.